PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule.
PROCEDURE: Not answered.
SPECIFIC CLINICAL QUESTION: Not answered.
OUTSIDE TISSUE DIAGNOSIS: Not answered.
PRIOR MALIGNANCY: Not answered.
CHEMORADIATION THERAPY: Not answered.
ORGAN TRANSPLANT: Not answered.
IMMUNOSUPPRESSION: Not answered.
OTHER DISEASES: Not answered.

UUID:A680FCC2-F8CD-445A-8523-C0A6857D52C5
TCGA-BJ-A0ZE-01A-PR Redacted

ICD-0-3
carcinoma, papillary, follicular variant 8340/3
Site: thyroid, NOS C73.9 lw 11/27/11

E-record history: Patient is a year-old female with prior history of left thyroid lobectomy. This is a completion thyroidectomy with a preop diagnosis of symptomatic thyroid goiter associated with a greater than 4-cm thyroid nodule.

ADDENDA:

Addendum

Criteria	Yes	No
Diagnosis Discrepancy		X
Primary Tumor Site Discrepancy		X
HIPAA Discrepancy		X
Prior Malignancy History		X
Dual/Synchronous Primary Noted		X
Case is (circle): QUALIFIED / DISQUALIFIED		
Reviewer Initials lw	Date Reviewed: 11/27/11	

lw 11/27/11

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block E:

A. **Mutations in *BRAF, NRAS61, HRAS61, KRAS12/13* NOT identified.**
B. **FISH test results are negative for *RET/PTC* rearrangement.**

NOTE:
DNA was extracted in the amount sufficient for testing.

BACKGROUND:
Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). *BRAF* V600E ( ) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between *BRAF* V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the *RAS* genes or *PAX8/PPARg* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, *BRAF* V600E mutation and *RET/PTC* and *PAX8/PPARg* rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). *RAS* mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. Am J Surg Pathol 2006, 30:216-222
2. Xing M, et al. BRAF mutation predicts a poorer clinical prognosis for papillary thyroid cancer. J Clin Endocrinol Metab 2005, 90:6373-6379.
3. Elisei R, et al. BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. J Clin Endocrinol Metab 2008, 93:3943-3949.
4. Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential BRAF(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. Ann Surg Oncol. 2009 Feb;16(2):240-5.
5. Nikiforova MN, et al. RAS point mutations and5PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. J Clin Endocrinol Metab 2003, 88: 2318-26.
6. Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV., ed. Endocrine Pathology: Differential Diagnosis and Molecular Advances. Humana Press, Totowa, 2004, 191-209.

MUTATIONAL ANALYSIS:
For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the platform to amplify BRAF codons 599-601, NRAS codon 61, HRAS codon 61, and KRAS codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by Sanger sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA.

FINAL DIAGNOSIS:

THYROID GLAND, RIGHT, LOBECTOMY (59 GRAMS) –

A. **PAPILLARY THYROID CARCINOMA, FOLLICULAR VARIANT, RIGHT LOBE (see comment).**
B. **SIZE: APPROXIMATELY 5.7 CM.**
C. **TUMOR EXTENDS TO MARGINS.**
D. **NO DEFINITE ANGIOLYMPHATIC INVASION OR EXTRATHYROIDAL EXTENSION.**
E. **MULTINODULAR THYROID.**
F. **NO PARATHYROID GLANDS ARE SEEN.**

[redacted]

[redacted]

COMMENT:
No definite angiolymphatic invasion or extrathyroidal extension are seen. Molecular studies will be performed and an addendum report will be issued.

MICROSCOPIC:

Microscopic examination substantiates the above diagnosis. Immunostains for HBME-1 was performed on block C, D, and E and shows positive focal membranous staining in the lesion.

The following statement applies to all immunohistochemistry, insitu hybridization (ISH & FISH), molecular anatomic pathology, and immunofluorescence testing:

The testing was developed and its performance characteristics determined by the l , as required by the CLIA '88 regulations. The testing has not been cleared or approved for the specific use by the U.S. Food and Drug Administration, but the FDA has determined such approval is not necessary for clinical use. Tissue fixation ranges from a minimum of 2 to a maximum of 84 hours.

This laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 ("CLIA") as qualified to perform high-complexity clinical testing. Pursuant to the requirements of CLIA, ASR's used in this laboratory have been established and verified for accuracy and precision. Additional information about this type of test is available upon request.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE:	Lobectomy
TUMOR SITE:	Right Lobe
TUMOR FOCALITY:	Unifocal
TUMOR SIZE (largest nodule):	Greatest Dimension: 5.7 cm
HISTOLOGIC TYPE:**	Papillary carcinoma, encapsulated follicular variant
PRIMARY TUMOR (pT):	pT3
REGIONAL LYMPH NODES (pN):	pNX
DISTANT METASTASIS (pM):	Not applicable
EXTRATHYROIDAL EXTENSION:	Not identified
MARGINS:	Margin(s) involved by carcinoma
LYMPH-VASCULAR INVASION:	Not identified
ADDITIONAL PATHOLOGIC FINDINGS:	Nodular goiter

SPECIAL PROCEDURES:

In Situ Procedure

Interpretation

RESULTS:

RET/PTC: The targeted area of the tissue showed 1(1.6%) of the cells with the rearrangement pattern and 61(98.4%) of the cells with the normal pattern. The targeted area is considered NOT rearranged for the *RET/PTC* region.

The *RET/PTC1* Probe did not contain the rearrangement pattern.

Results

RET/PTC Methodology Description:

Probe: *RET/PTC* Probe (10q11.2) and *RET/PTC1* Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The *RET* probe (labeled SpectrumGreen) is a 207 kb DNA probe located at band 10q11.2.

The H4 and 369 probes are approximately 100kb each (labeled in), and are located at band 10q21.

RET/PTC rearrangement is suggested to be present when three RET signals are identified in greater than 8% of the analyzed cells.

RET /PTC1 rearrangement detection is as follows: Normal cells without the *RET/PTC1* rearrangement show two pair of green and orange signals separated. Cells with the *RET/PTC1* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the *PTC* (369/H4) SpectrumOrange and the *RET* SpectrumGreen probes.